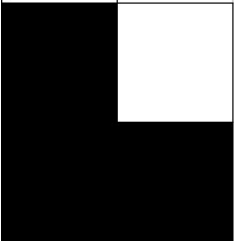


DTL Researcher Test

1. (16p) Let x be an integer satisfying $1 \leq x \leq 10^{100}$, what is the probability that x^3 ends with 11?
2. (16p) Suppose that a, b, c, d are positive real numbers satisfying $(a + c)(b + d) = ac + bd$.
Find the smallest possible value of $S = \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}$.
3. (16p) Let $a_1, a_2, \dots, a_{2021}$ be positive integers. Prove that there exist at least two different sequences $\{a_i\}$, $i = 1, 2, \dots, 2021$, such that $\frac{1}{a_1} + \frac{1}{2a_2} + \dots + \frac{1}{2021a_{2021}} = 1$.
4. (20p) We define the figure composed of any three squares in 2×2 squares as L-shaped, see the right figure for example.


(1) Suppose there is a chessboard with $2^n \times 2^n$, $n \in \mathbb{N}$, squares. At first, we put a square board on it covering exactly one square, then we put many L-shaped boards (each can cover 3 squares exactly) over the rest of the chessboard. Prove that for any positive integer n and any position of the first square board, we can cover the whole chessboard without non-overlapping boards.

(2) Suppose there is an 8×8 chessboard, we can put L-shaped boards (each board can cover exactly 3 squares) into chessboard without overlapping. How many L-shaped boards should we put at least, then there is no more space for another non-overlapping L-shaped board on the chessboard?
5. (16p) There is a number set with three numbers: $2, \sqrt{2}$ and $1/\sqrt{2}$. In each turn, you can choose any two numbers, (denoted as a, b) among them and replace a, b with $(a+b)/\sqrt{2}$ and $(a-b)/\sqrt{2}$. For example, if you chose $2, \sqrt{2}$ at first turn, then the set becomes: $(2+\sqrt{2})/\sqrt{2}, (2-\sqrt{2})/\sqrt{2}$ and $1/\sqrt{2}$. Then you can replace two numbers from this new number set. Can we change the number set to $1, \sqrt{2}$ and $1 + \sqrt{2}$ in finite steps? Give all steps needed if we can, or justification otherwise.
6. (16p) Each of eight boxes contains six balls. Each ball has been colored with one of n colors, such that no two balls in the same box are the same color, and no two colors occur together in more than one box. Determine, with justification, the smallest integer n for which this is possible.